

Question 1:-

AR(2): $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + W_t$ W_t is white noise (casual)

Multiply each side by X_{t-h} for $h > 0$ & take expectation.

$$E[X_t X_{t-h}] = \phi_1 E[X_{t-1} X_{t-h}] + \phi_2 E[X_{t-2} X_{t-h}] + E(W_t X_{t-h})$$

$$\gamma(h) = \phi_1 \gamma(h-1) + \phi_2 \gamma(h-2), \quad h=1, 2, \dots$$

Now, we will divide it by $\gamma(0)$

$$f(h) = \phi_1 f(h-1) + \phi_2 f(h-2), \quad h \geq 2$$

initial condition: $f(0) = 1$ $f(1) = \phi_1 + \phi_2 f(0)$

$$(1 - \phi_2) f(1) = \phi_1 \quad (\because f(-1) = f(0))$$

$$\therefore f(1) = \frac{\phi_1}{1 - \phi_2}$$

$$(1 - \phi_1 B - \phi_2 B^2) X_t = W_t$$

General Solution of $1 - \phi_1 z - \phi_2 z^2 = 0$

Case ①:- If distinct roots z_1, z_2 then $f(h) = c_1 z_1^{-h} + c_2 z_2^{-h}$

Case ②:- If both roots equal then $f(h) = z_0^{-h} (c_1 + c_2 h)$

Case ③:- Complex roots $z = r e^{\pm i\theta}$.

$$f(h) = A r^{-h} \cos(h\theta + \phi).$$

① $X_t + 1.6 X_{t-1} + 0.64 X_{t-2} = W_t$

$$\phi_1 = -1.6$$

$$\phi_2 = -0.64$$

$$f(0) = 1$$

$$f(1) = \frac{\phi_1}{1 - \phi_2} = \frac{-1.6}{1 - (-0.64)}$$

$$f(1) = 0.976$$

from the characteristic eqⁿ:-

$$\therefore f(1) = \frac{-40}{41} \approx 0.976$$

$$1 + 1.6z + 0.64z^2 = 0$$

$$64z^2 + 160z + 100 = 0$$

$$(8z + 10)^2 = 0 \Rightarrow z_1 = z_2 = -\frac{10}{8} = -1.25 = z_0$$

$$\therefore f(h) = z_0^{-h} (c_1 + c_2 h)$$

$$\therefore f(0) = 1 \Rightarrow 1 = c_1$$

$$\underline{h=2}; f(1) = \frac{1+c_2}{z_0} = -\frac{40}{41}$$

$$c_2 = \left(-\frac{40}{41} \times (-1.25) \right) - 1$$

$$c_2 = \frac{9}{41}$$

$$f(h) = (-1.25)^{-h} \left(1 + \frac{9}{41} h \right)$$

ACF values till lag 10 (All values are rounded off)

lag h	f(h)	lag (h)	f(h)
0	1	6	0.61
1	-0.976	7	-0.53
2	0.921	8	0.46
3	-0.85	9	-0.40
4	0.77	10	0.34
5	-0.69		

(b) $X_t - 1.2X_{t-1} + 0.85X_{t-2} = W_t$

$$\phi_1 = 1.2 \quad \phi_2 = -0.85$$

initial condⁿ $f(h) = 1.2f(h-1) - 0.85f(h-2), h \geq 2$

$$f(0) = 1, f(1) = \frac{\phi_1}{1 - \phi_2} = \frac{1.2}{1.85} \approx 0.649 \quad (\text{rounded off till 3 digits decimals})$$

characteristic eqⁿ: $0.85z^2 - 1.2z + 1 = 0$

$$z^2 - \frac{1.2}{0.85}z + \frac{1}{0.85} = 0$$

$$z = \frac{\frac{1.2}{0.85} \pm \sqrt{\left(\frac{1.2}{0.85}\right)^2 - 4 \cdot \left(\frac{1}{0.85}\right)}}{2}$$

$$\therefore \boxed{z = 0.706 \pm 0.824i}$$

$$|z_1| = \sqrt{(0.706)^2 + (0.824)^2} = 1.085$$

$$\theta = \tan^{-1}\left(\frac{0.824}{0.706}\right) \approx \tan^{-1}(1.167) \approx 0.862 \text{ radians}$$

Hence,

$$f(h) = A|z_1|^{-h} \cos(h\theta + b)$$

$$\boxed{A \approx 1.002}$$

$$\boxed{b \approx -0.069}$$

$$\therefore \boxed{f(h) \approx 1.002 (1.085)^{-h} \cos(0.862h - 0.069)}$$

$\log(h)$	$f(h)$		$\log(h)$	$f(h)$
0	1		6	0.235
1	0.649		7	0.540
2	-0.072		8	0.447
3	-0.637		9	0.079
4	-0.704		10	-0.286
5	-0.303			

MTH442: Assignment 3

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Question 1

Part (a)

The model is $X_t + 1.6X_{t-1} + 0.64X_{t-2} = W_t$.

So the AR coefficients are $\phi_1 = -1.6$ and $\phi_2 = -0.64$.

The recurrence is $\rho(h) = -1.6\rho(h-1) - 0.64\rho(h-2)$ with $\rho(0)=1$ and $\rho(1) = \phi_1/(1-\phi_2)$.

I will now check with ARMAacf.

```
library(stats)
phi <- c(-1.6, -0.64)
h <- 10
rho_arma <- ARMAacf(ar=phi, ma = numeric(0), lag.max = h)
data.frame(lag=0:h, arma = round(rho_arma,10))
```

```
##      lag      arma
## 0      0 1.0000000
## 1      1 -0.9756098
## 2      2 0.9209756
## 3      3 -0.8491707
## 4      4 0.7692488
## 5      5 -0.6873288
## 6      6 0.6074068
## 7      7 -0.5319605
## 8      8 0.4623964
## 9      9 -0.3993796
## 10     10 0.3430736
```

Part (b)

The model is $X_t - 1.2X_{t-1} + 0.85X_{t-2} = W_t$.

So the AR coefficients are $\phi_1 = 1.2$ and $\phi_2 = -0.85$.

The recurrence is $\rho(h) = 1.2\rho(h-1) - 0.85\rho(h-2)$ with $\rho(0)=1$ and $\rho(1)=\phi_1/(1-\phi_2)$.

Now, check part(B) with ARMAacf.

```
phi <- c(1.2, -0.85)
h <- 10
rho_arma <- ARMAacf(ar=phi, ma = numeric(0), lag.max = h)
data.frame(lag=0:h, arma=round(rho_arma,10))
```

```
##      lag      arma
## 0      0  1.00000000
## 1      1  0.64864865
## 2      2 -0.07162162
## 3      3 -0.63729730
## 4      4 -0.70387838
## 5      5 -0.30295135
## 6      6  0.23475500
## 7      7  0.53921465
## 8      8  0.44751583
## 9      9  0.07868654
## 10     10 -0.28596460
```

Question 2

Part 1: Global Land Temperatures

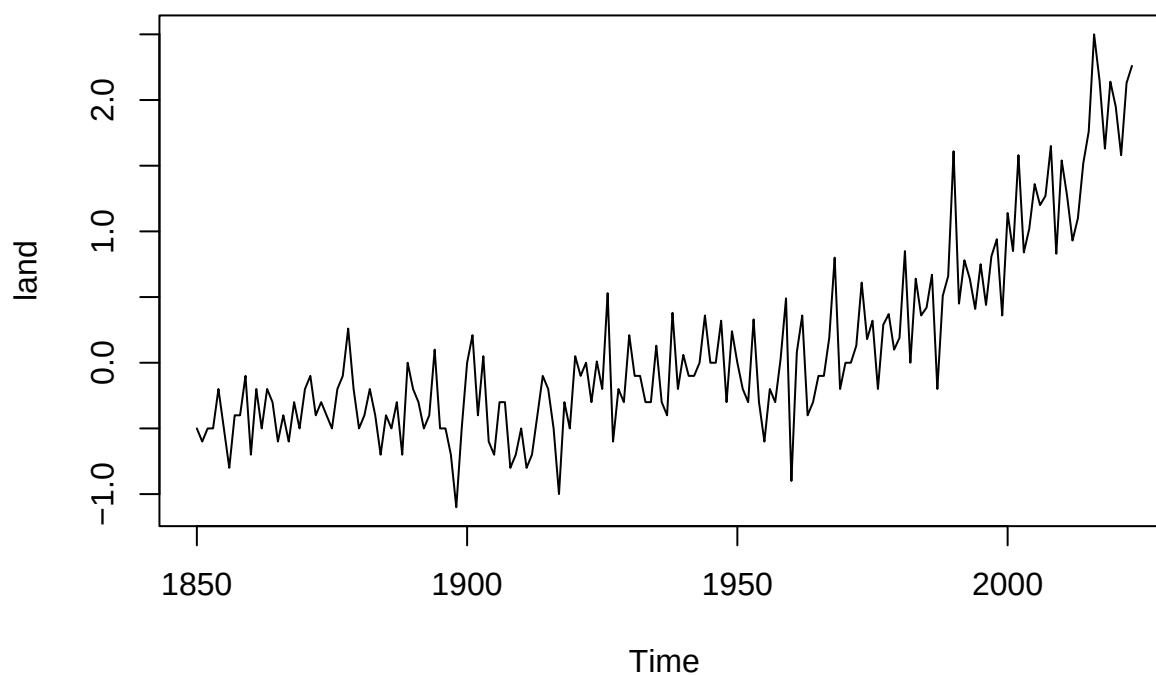
Load Libraries

```
library(astsa)
library(forecast)
```

Now, We will load the gtemp_land data and plot it to see what it looks like

```
data(gtemp_land)
land <- gtemp_land
plot(land, main="Global Land Temperatures")
```

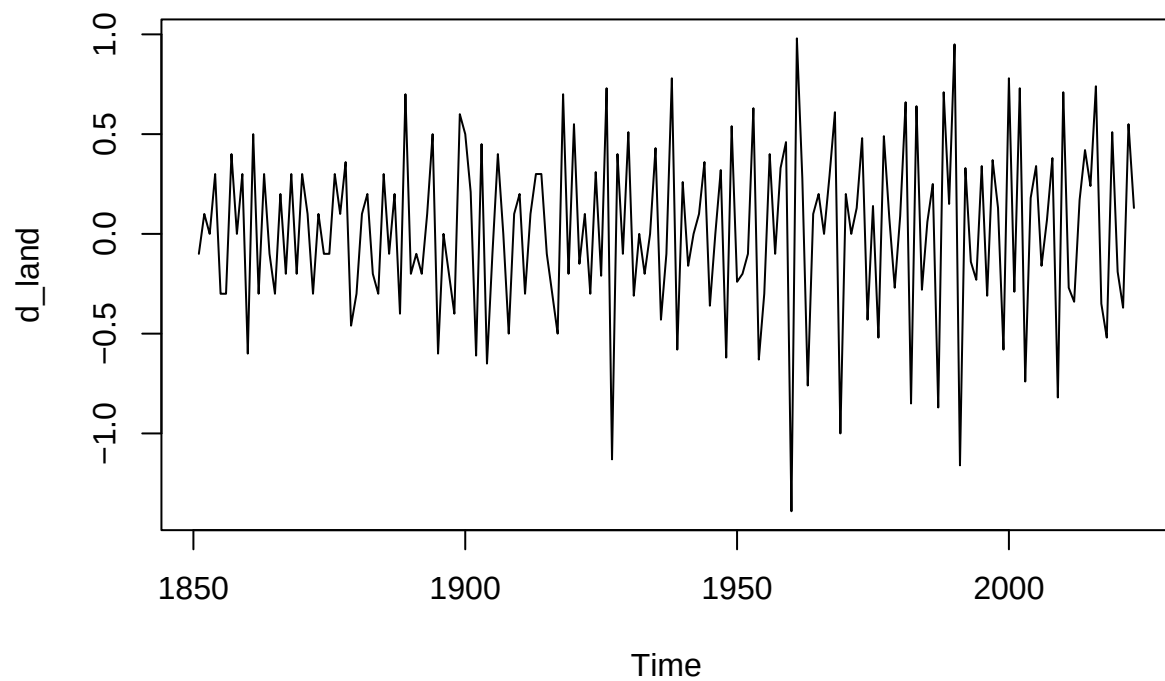
Global Land Temperatures



The plot clearly shows an upward trend, so the data isn't stationary. This means I need to use differencing. I'll start with $d=1$

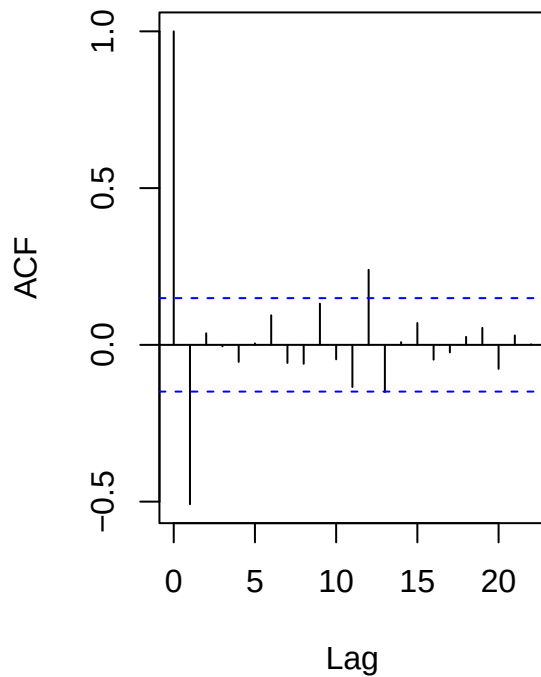
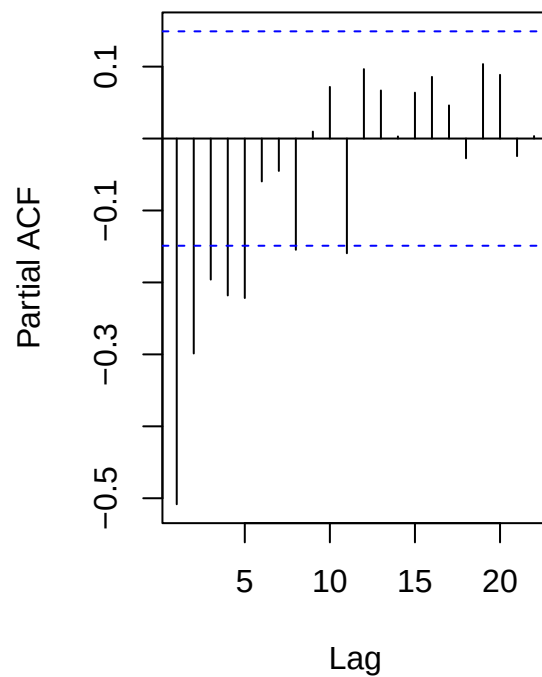
```
d_land <- diff(land)
plot(d_land, main="Differenced Land Temperatures")
```

Differenced Land Temperatures



This plot of the differenced data looks much more stationary. It's centered around a mean value. Now I'll look at the ACF and PACF plots to get an idea for p and q.

```
par(mfrow = c(1, 2))
acf(d_land, main="ACF of Differenced Land Data")
pacf(d_land, main="PACF of Differenced Land Data")
```

ACF of Differenced Land Data**PACF of Differenced Land Data**

The ACF and PACF plots are a bit hard to read. Both seem to have a few significant lags that trail off.

To find a good model, I'll just use the `auto.arima` function. It will automatically check many different models and pick the one with the best (lowest) AICc score.

```
best_land <- auto.arima(land)
best_land
```

```
## Series: land
## ARIMA(0,1,1) with drift
##
## Coefficients:
##          ma1    drift
##        -0.7974  0.0145
## s.e.    0.0395  0.0052
##
## sigma^2 = 0.1104: log likelihood = -54.39
## AIC=114.78  AICc=114.93  BIC=124.24
```

The function chose an ARIMA(0, 1, 1) with drift. This means:

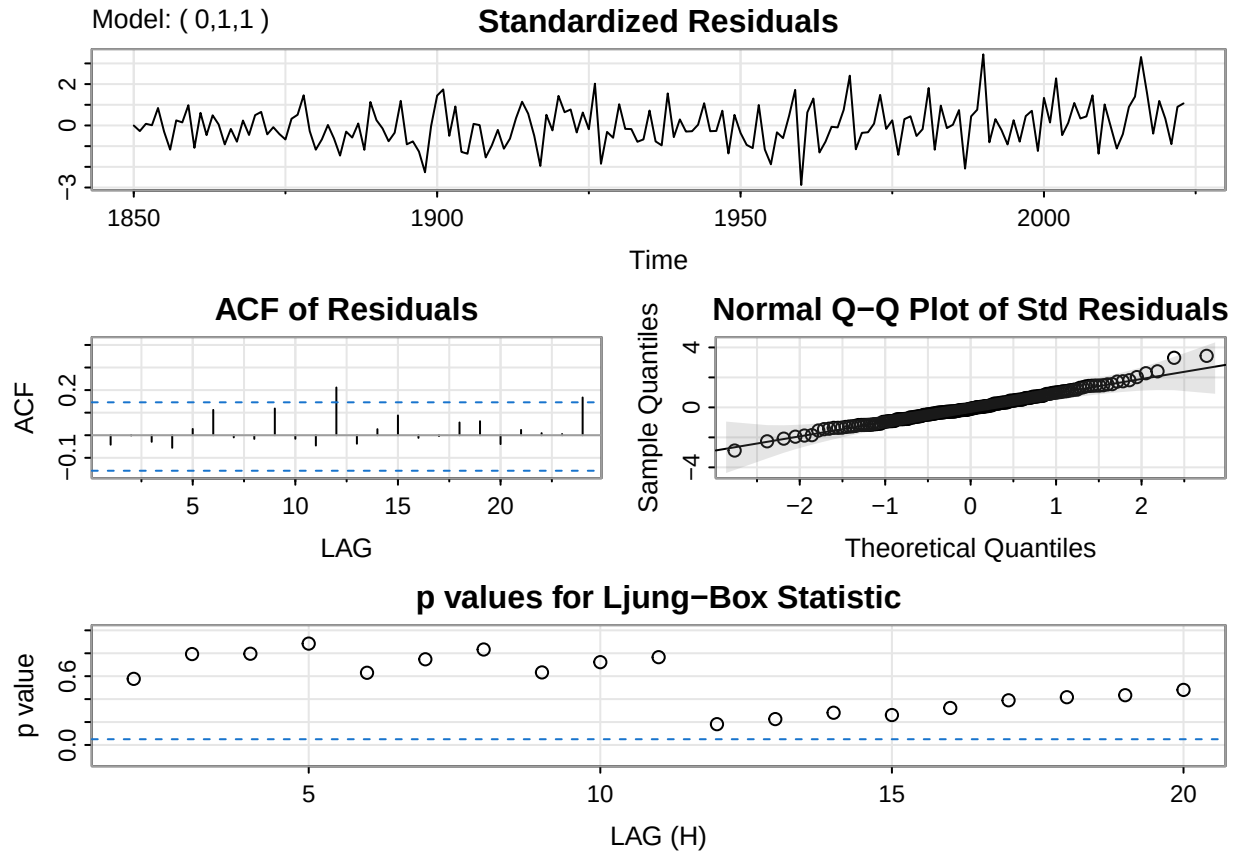
$p = 0$ (AR part), $d = 1$ (Differencing), $q = 1$ (MA part)

The “drift” term is included to model the overall upward trend.

Now I'll fit this model using the `sarima` function to check the diagnostics.

```
fit_land <- sarima(land, p = 0, d = 1, q = 1)
```

```
## initial  value -0.842449
## iter    2 value -1.043995
## iter    3 value -1.086437
## iter    4 value -1.105162
## iter    5 value -1.106533
## iter    6 value -1.107112
## iter    7 value -1.107288
## iter    8 value -1.107352
## iter    9 value -1.107368
## iter   10 value -1.107368
## iter   10 value -1.107368
## iter   10 value -1.107368
## final   value -1.107368
## converged
## initial  value -1.104513
## iter    2 value -1.104517
## iter    3 value -1.104520
## iter    4 value -1.104533
## iter    4 value -1.104533
## iter    4 value -1.104533
## final   value -1.104533
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE  t.value p.value
## ma1          -0.7974 0.0395 -20.1909  0.0000
## constant     0.0145 0.0052   2.7905  0.0059
##
## sigma^2 estimated as 0.1091637 on 171 degrees of freedom
##
## AIC = 0.663493  AICc = 0.663901  BIC = 0.7181744
##
```



The diagnostic plots from sarima look good.

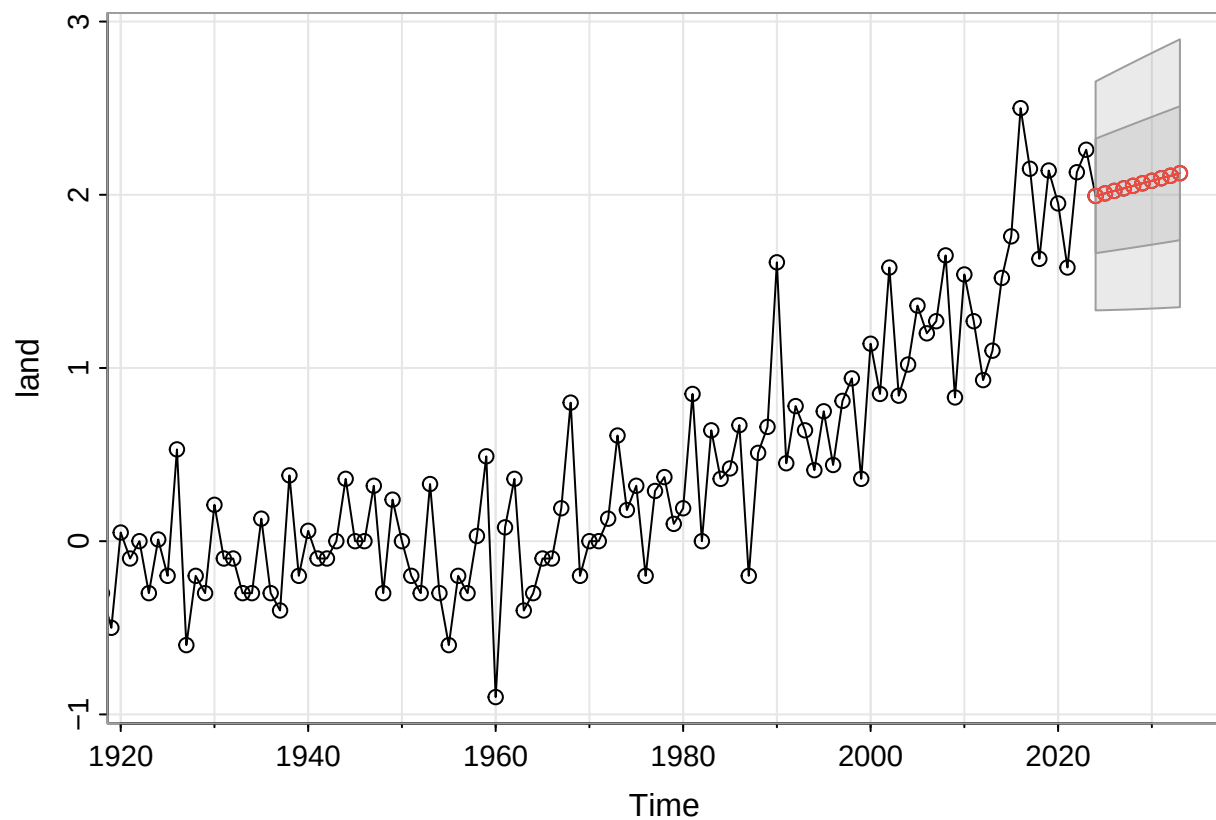
The “Standardized Residuals” look random and centered around zero.

The “ACF of Residuals” plot shows that almost no lags are significant. This means the residuals aren’t correlated, which is what I want.

The “Ljung-Box” p-values (the circles) are all high, above the dotted line (0.05). This means the residuals are behaving like random white noise.

The model seems to be a good fit. Now I will forecast the next 10 years.

```
for_land <- sarima.for(land, n.ahead = 10, p = 0, d = 1, q = 1)
```

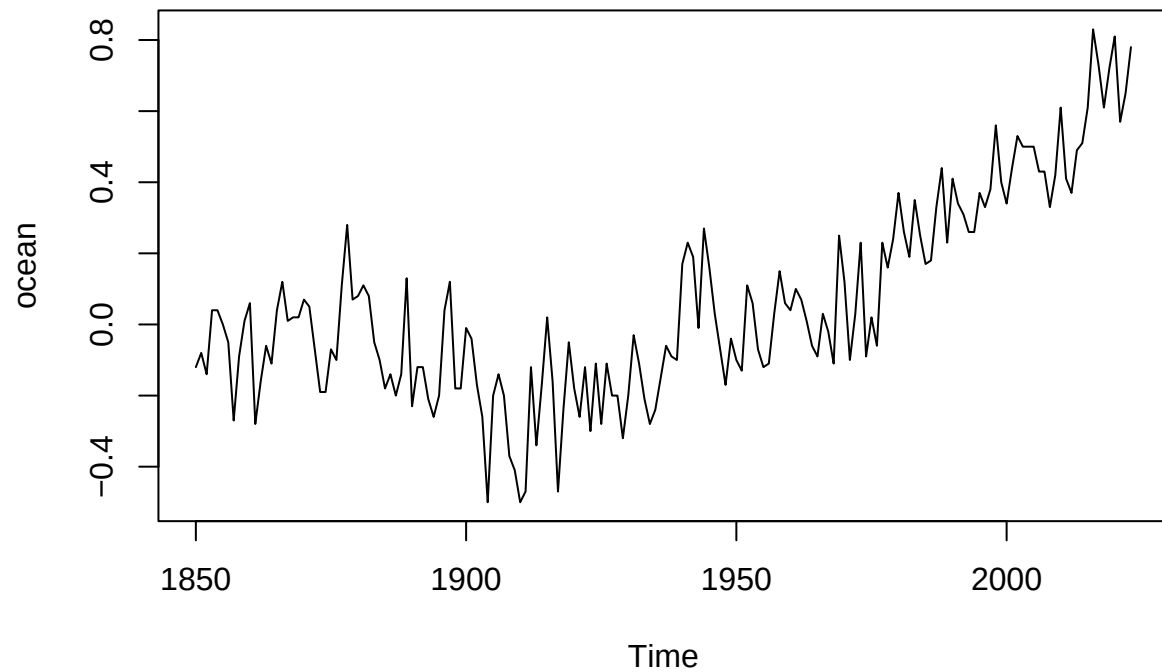


The forecast shows that the global land temperature is predicted to continue rising. The prediction intervals (the 'L' and 'U' columns) get wider as I forecast further into the future, which makes sense because forecasts become less certain over time.

Part 2: Global Ocean Temperatures

```
data(gtemp_ocean)
ocean <- gtemp_ocean
plot(ocean, main="Global Ocean Temperatures")
```

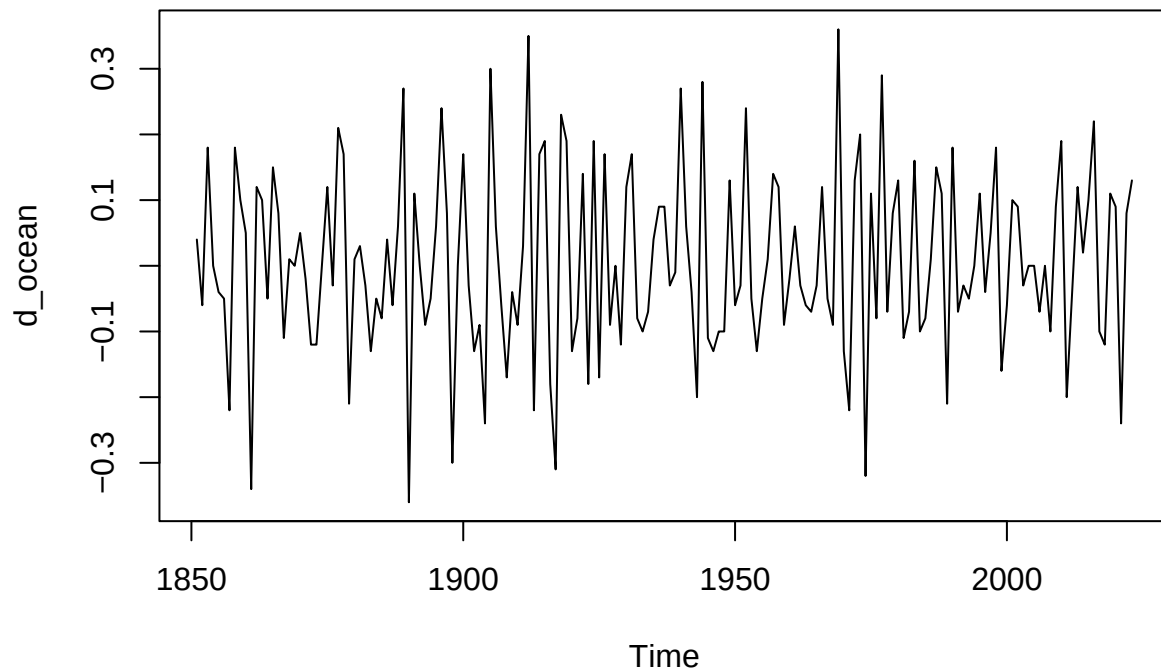
Global Ocean Temperatures



This data also has a clear upward trend, just like the land data. So I'll use $d=1$ again.

```
d_ocean <- diff(ocean)
plot(d_ocean, main="Differenced Ocean Temperatures")
```

Differenced Ocean Temperatures



The differenced data looks stationary. Now I'll use `auto.arima` to find the best model.

```
best_ocean <- auto.arima(ocean)
best_ocean
```

```
## Series: ocean
## ARIMA(0,1,2) with drift
##
## Coefficients:
##          ma1          ma2      drift
##        -0.4256   -0.3093   0.0045
## s.e.    0.0728    0.0715   0.0025
##
## sigma^2 = 0.01516:  log likelihood = 118.03
## AIC=-228.06   AICc=-227.82   BIC=-215.45
```

For the ocean data, `auto.arima` picked an `ARIMA(0, 1, 2)` with drift. This means:

$p = 0$ (no AR part), $d = 1$ (Differencing), $q = 2$ (MA part)

This seems like a common model for data with a simple trend. I'll fit it and check the diagnostics.

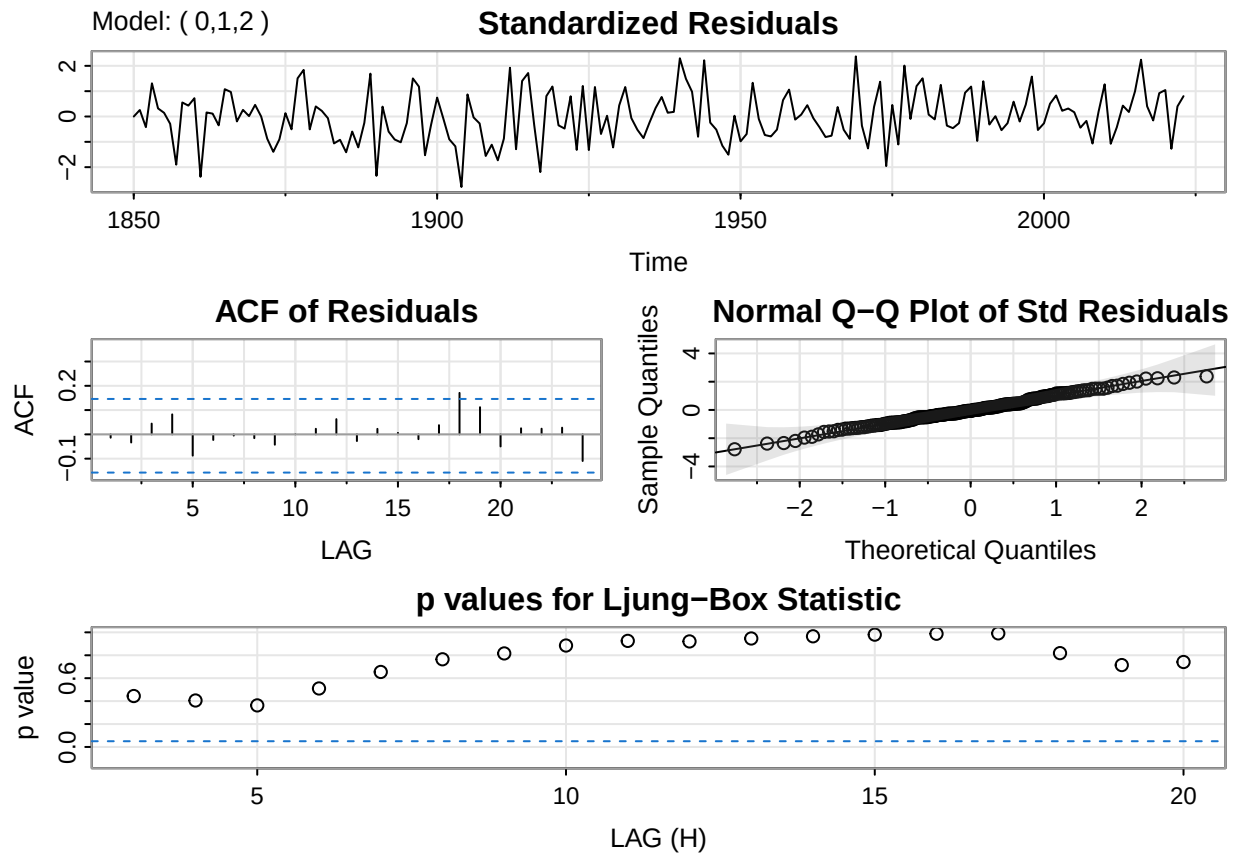
```
fit_ocean <- sarima(ocean, p = 0, d = 1, q = 2)
```

```
## initial value -1.970301
## iter 2 value -2.078881
```

```

## iter    3 value -2.097771
## iter    4 value -2.100283
## iter    5 value -2.101226
## iter    6 value -2.102299
## iter    7 value -2.102967
## iter    8 value -2.102974
## iter    9 value -2.102974
## iter   10 value -2.102974
## iter   11 value -2.102974
## iter   11 value -2.102974
## iter   11 value -2.102974
## final   value -2.102974
## converged
## initial  value -2.101175
## iter    2 value -2.101193
## iter    3 value -2.101198
## iter    4 value -2.101198
## iter    4 value -2.101198
## iter    4 value -2.101198
## final   value -2.101198
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ma1          -0.4256 0.0728 -5.8472  0.0000
## ma2          -0.3093 0.0715 -4.3233  0.0000
## constant      0.0045 0.0025  1.8006  0.0735
##
## sigma^2 estimated as 0.01490112 on 170 degrees of freedom
##
## AIC = -1.318277  AICc = -1.317456  BIC = -1.245368
##

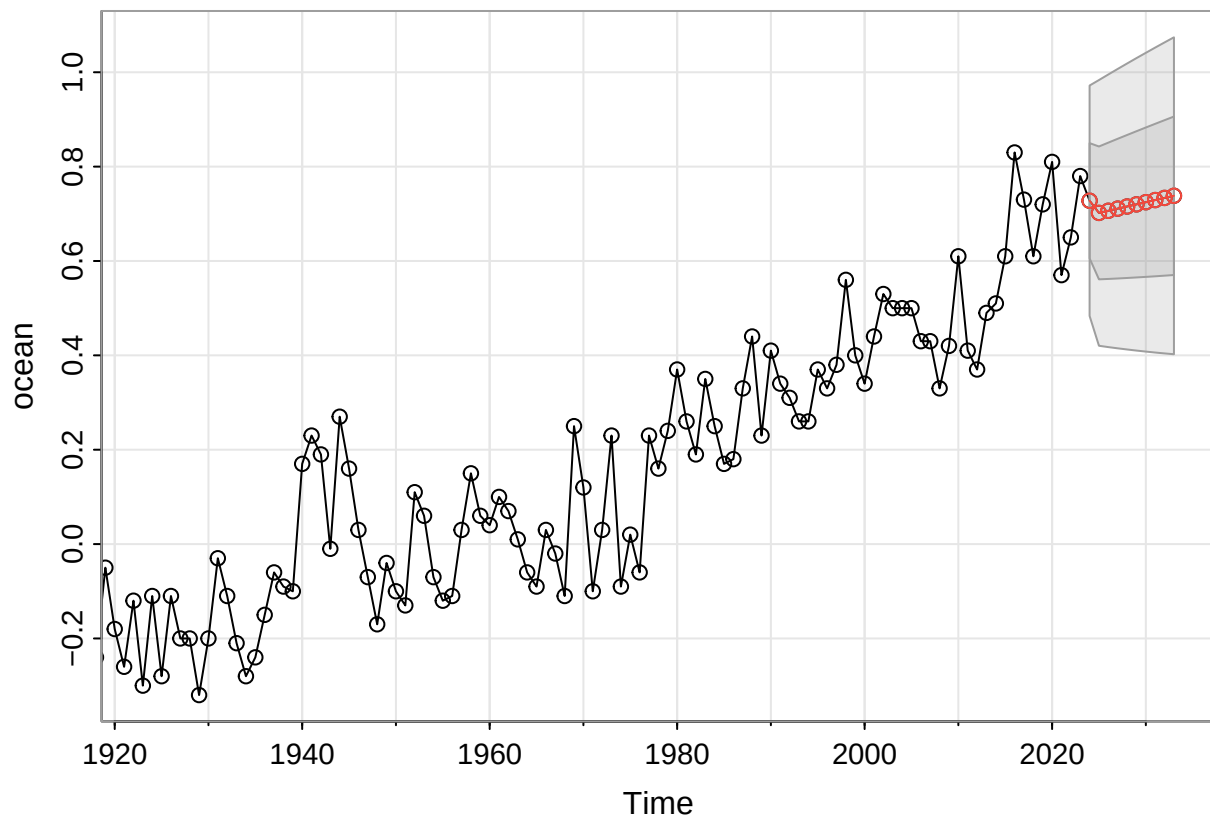
```



These diagnostics also look very good. The residuals are random, their ACF is clean (no significant lags), and the Ljung-Box p-values are all high. This model is a good fit for the ocean data.

Now, I'll make the 10-year forecast.

```
for_ocean <- sarima.for(ocean, n.ahead = 10, p = 0, d = 1, q = 2)
```



Comments

The forecast for the ocean temperatures also shows a clear warming trend. The models for both the land and ocean data are very similar (ARIMA(0,1,1) and ARIMA(0,1,2)). Both models required one difference ($d=1$) and a drift term to account for the strong upward trend in global temperatures. My forecasts for both datasets predict this warming will continue for the next 10 years.

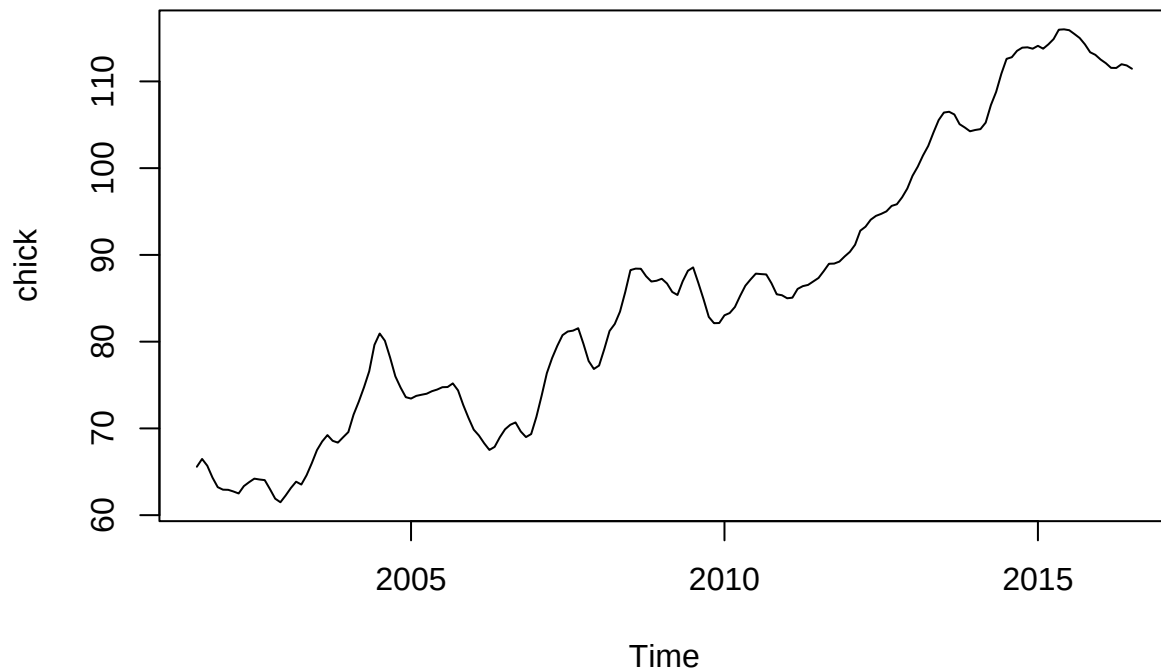
Question 3

First, I'll load the chicken data and plot it to see what I'm working with.

Part 1: Chicken Price Data

```
data(chicken)
chick <- chicken
plot(chick, main="Chicken Prices")
```

Chicken Prices



The plot shows a clear upward trend and also a very strong seasonal pattern. Both of these mean the data is not stationary.

Because it's hard to guess all the p, d, q and P, D, Q parts just from looking, I'll use the `auto.arima` function to find a good starting model. This function will automatically find the best values for d and D (differencing) and a good model to fit.

```
best_chick <- auto.arima(chick)
best_chick
```

```
## Series: chick
## ARIMA(2,1,1)(0,0,1)[12] with drift
##
## Coefficients:
##          ar1      ar2      ma1      sma1      drift
##          1.2933 -0.5375 -0.4019  0.2756  0.2518
## s.e.    0.2220  0.1542  0.2569  0.0692  0.1428
##
## sigma^2 = 0.396: log likelihood = -169.51
## AIC=351.01  AICc=351.5  BIC=370.14
```

`auto.arima` picked an `ARIMA(2,1,1)(0,0,1)[12]` with drift. This means:

p=2, d=1, q=1 (non-seasonal part)

P=0, D=0, Q=1 (seasonal part)

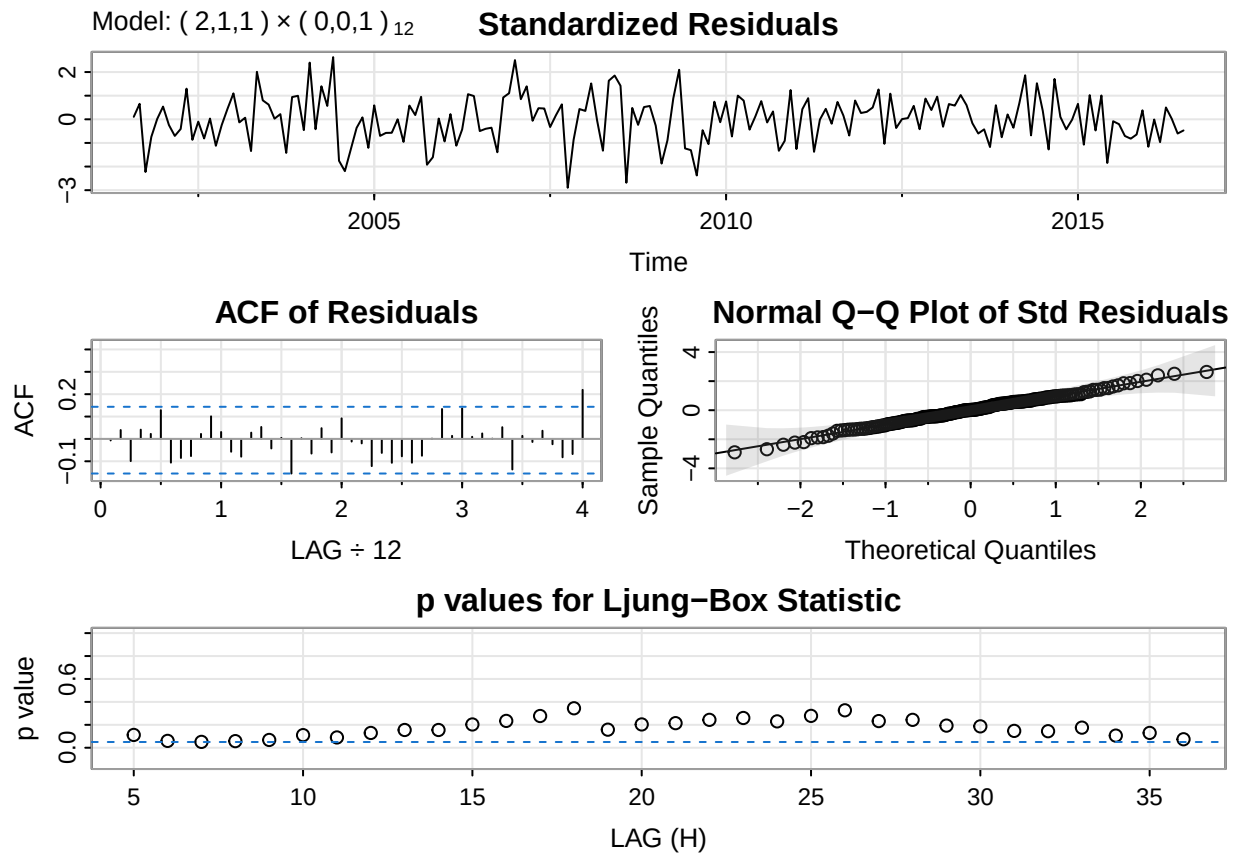
S=12 (seasonal period is 12 months)

D=0 means it didn't need a full seasonal difference, the seasonal MA term Q=1 must be enough to handle the seasonality.

```
fit_chick <- sarima(chick, p = 2, d = 1, q = 1, P = 0, D = 0, Q = 1, S = 12)
```

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```
## ar2      -0.5375 0.1542 -3.4849 0.0006
## ma1      -0.4019 0.2569 -1.5647 0.1195
## sma1      0.2756 0.0692 3.9809 0.0001
## constant 0.2518 0.1428 1.7641 0.0795
##
## sigma^2 estimated as 0.3849259 on 174 degrees of freedom
##
## AIC = 1.960969 AICc = 1.962906 BIC = 2.067808
##
```



The diagnostic plots look good.

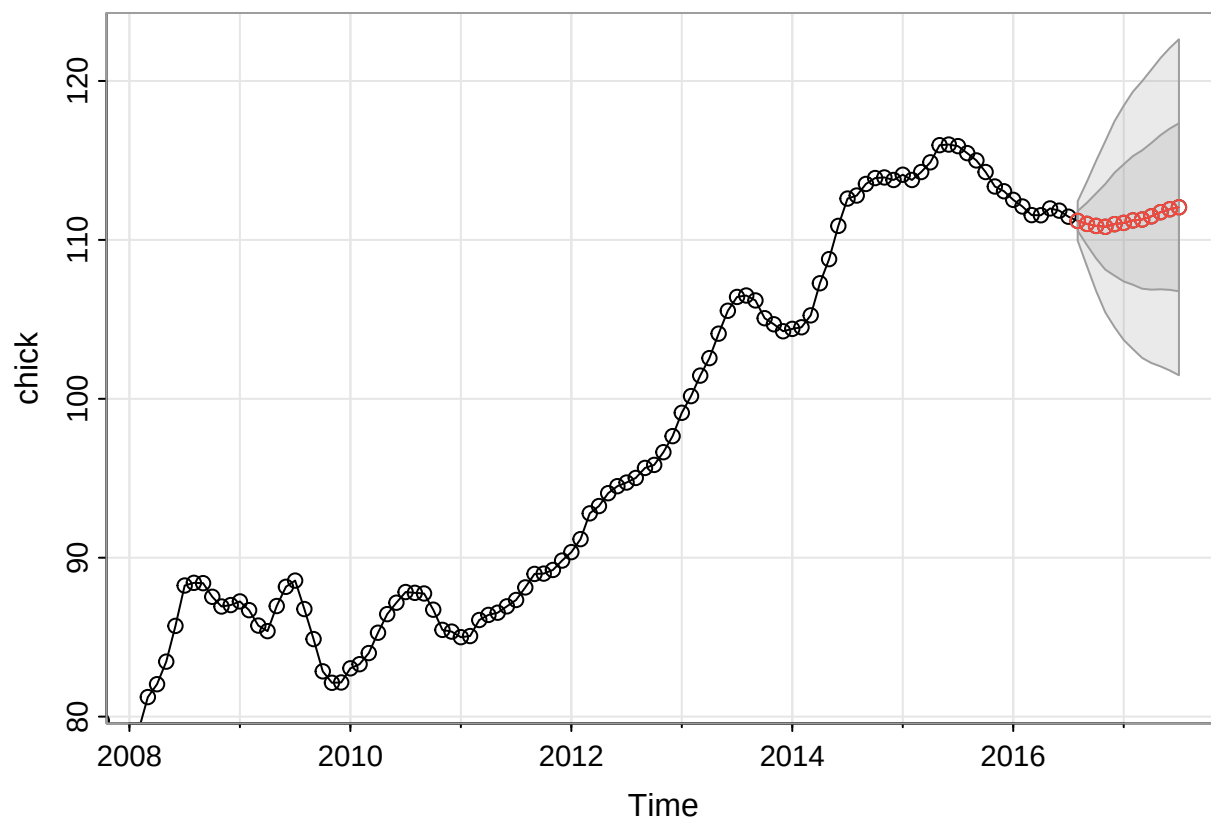
The “Standardized Residuals” plot looks random and flat.

The “ACF of Residuals” plot shows no significant spikes. This means the model captured all the correlation.

The “Ljung-Box” p-values are all high (above the 0.05 line), which confirms the residuals are random white noise.

This model seems to be a good fit. Now I’ll use it to forecast the next 12 months.

```
for_chick <- sarima.for(chick, n.ahead = 12, p = 2, d = 1, q = 1, P = 0, D = 0, Q = 1, S = 12)
```



The forecast continues the upward trend and also shows the seasonal pattern. The forecast limits (the shaded area) get wider, which makes sense.

Comments:

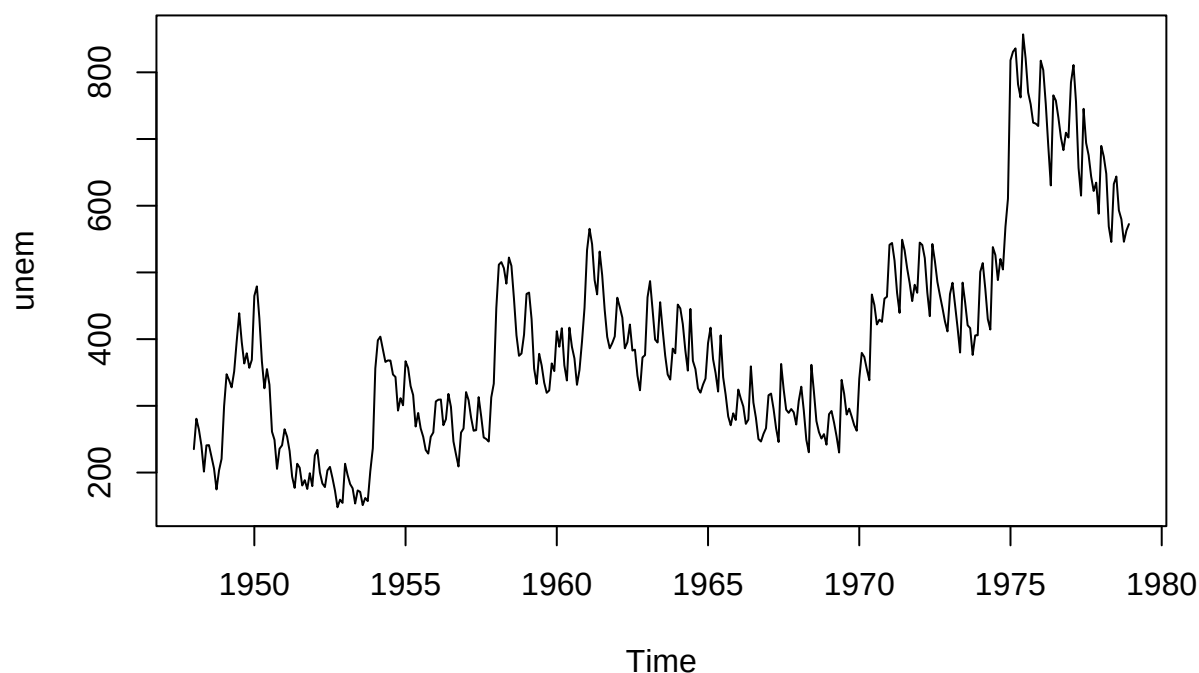
The SARIMA(2,1,1)(0,0,1)[12] proved to be a good fit for the chicken data. The diagnostics were clean, with random residuals and high Ljung-Box p-values. This SARIMA model captured both the strong upward trend and the monthly seasonality. My 12-month forecast reflects this, showing a continued rise in prices while following the expected seasonal pattern.

Part 2: Unemployment Data (unemp)

Now I'll redo the whole analysis for the unemp dataset.

```
data(unemp)
unem <- unemp
plot(unem, main="Unemployment Data")
```

Unemployment Data



This data looks different. It has seasonal patterns, but the trend isn't a simple straight line. It wanders up and down.

I'll use `auto.arima` again to see what model it suggests.

```
best_unem <- auto.arima(unem)
best_unem
```

```
## Series: unem
## ARIMA(5,0,0)(2,1,1)[12] with drift
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      sar1      sar2      sma1
##      1.1093  0.0739 -0.2046  0.0959 -0.1147 -0.1796 -0.1278 -0.5318
## s.e.  0.0530  0.0806  0.0782  0.0794  0.0529  0.1155  0.0889  0.1076
##      drift
##      1.1571
## s.e.  0.8226
##
## sigma^2 = 437.9: log likelihood = -1606.32
## AIC=3232.64  AICc=3233.27  BIC=3271.5
```

`auto.arima` chose an `ARIMA(5,0,0)(2,1,1)[12]` model. This means:

$p=5$, $d=0$, $q=0$ (non-seasonal part)

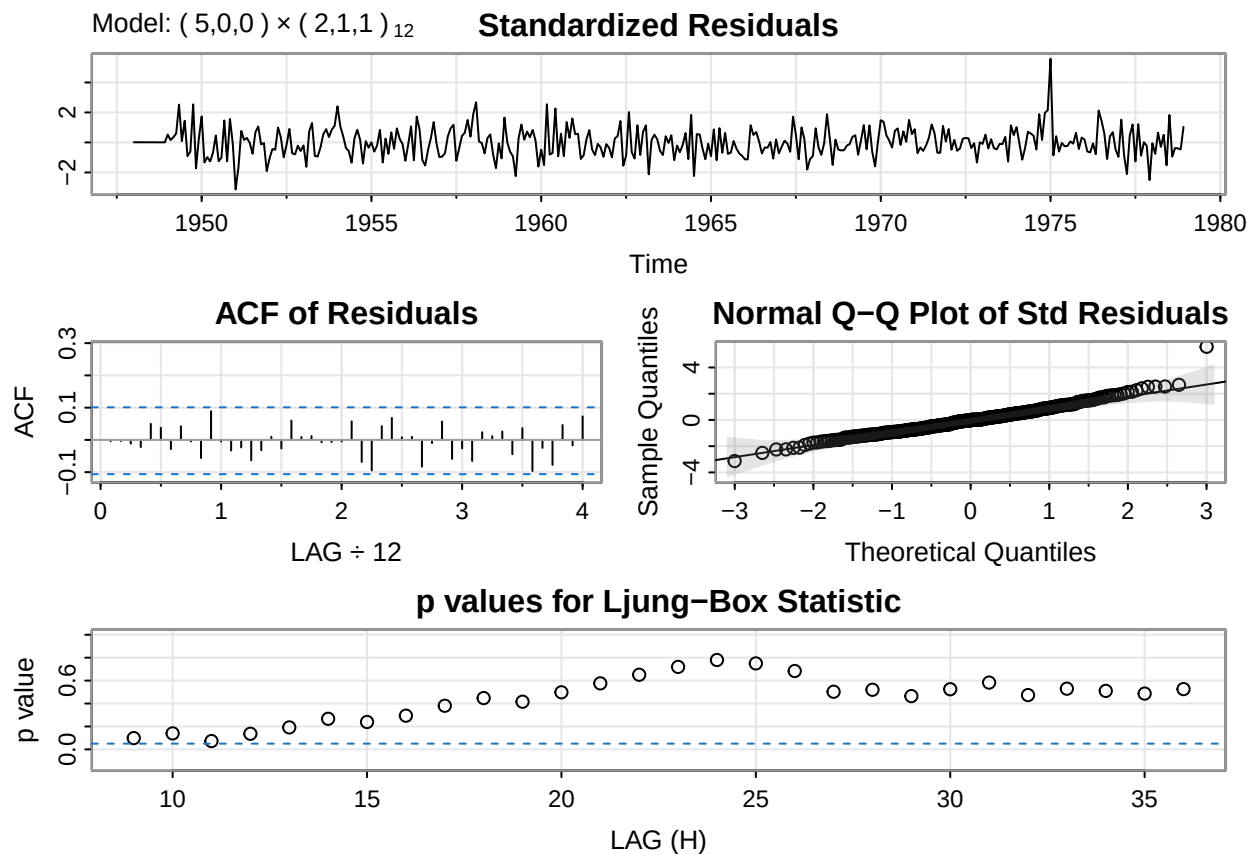
$P=2$, $D=1$, $Q=1$ (seasonal part)

This is a more complex model. It uses D=1 for the seasonality and a high-order AR(5) for the non-seasonal part to model the wandering behavior. It also includes a drift term.

```
fit_unem <- sarima(unem, p = 5, d = 0, q = 0, P = 2, D = 1, Q = 1, S = 12)
```

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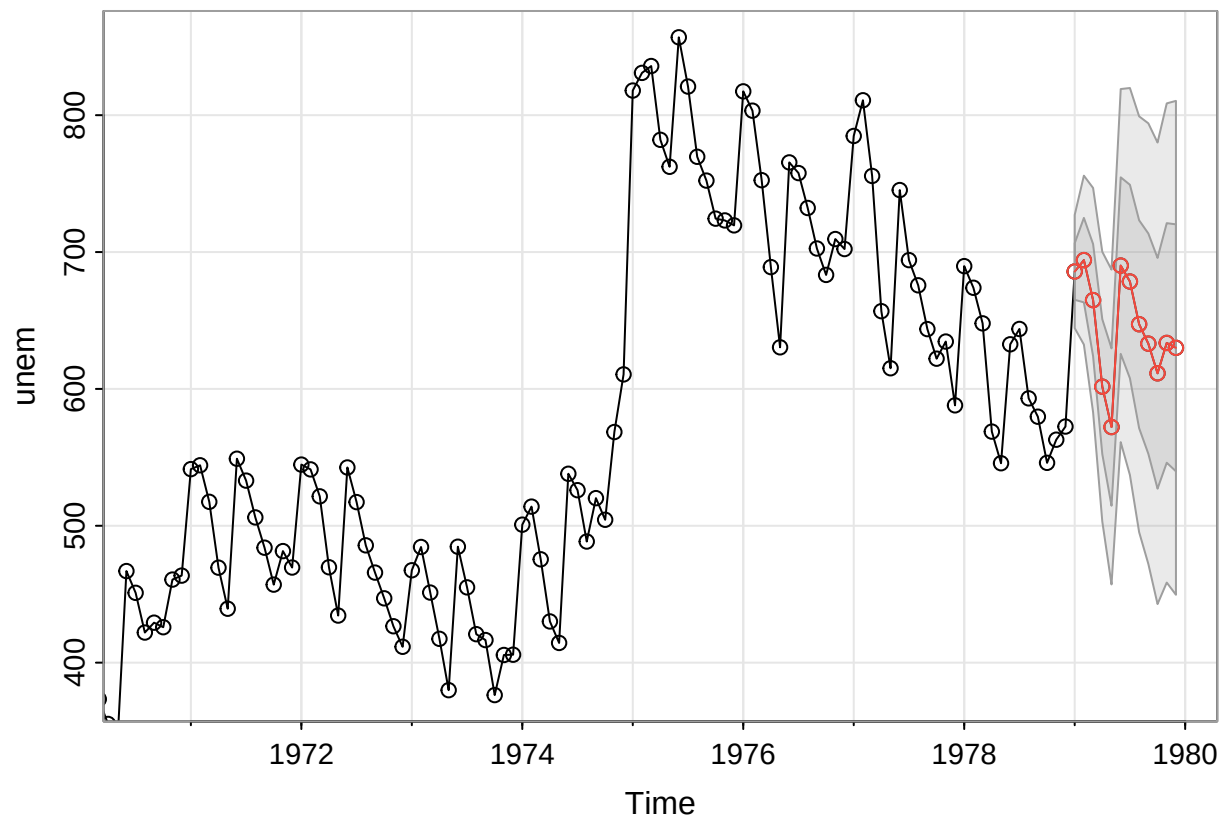
```
## ar3      -0.2046 0.0782 -2.6156 0.0093
## ar4      0.0959 0.0794 1.2076 0.2280
## ar5     -0.1147 0.0529 -2.1678 0.0308
## sar1     -0.1796 0.1155 -1.5557 0.1207
## sar2     -0.1278 0.0889 -1.4377 0.1514
## sma1     -0.5318 0.1076 -4.9415 0.0000
## constant 1.1571 0.8226 1.4066 0.1604
##
## sigma^2 estimated as 426.9507 on 351 degrees of freedom
##
## AIC = 8.979558 AICc = 8.980987 BIC = 9.087506
##
```



These diagnostics also look good. The residuals are random, the ACF has no big spikes, and the Ljung-Box p-values are all high. This model seems to fit well.

Finally, I'll forecast the next 12 months.

```
for_unem <- sarima.for(unem, n.ahead = 12, p = 5, d = 0, q = 0, P = 2, D = 1, Q = 1, S = 12)
```



The forecast shows the seasonal pattern continuing. This model seems to predict the unemployment level will stay somewhat steady but continue its seasonal cycle.

Comments:

For the unemployment data, I used the SARIMA(5,0,0)(2,1,1)[12]. This complex SARIMA model was needed to capture the data's wandering trend and strong seasonality. The diagnostics were again very good, confirming the fit. The 12-month forecast shows the seasonal pattern continuing at the more recent, stable level.